Real-world Anomaly Detection in Surveillance Videos

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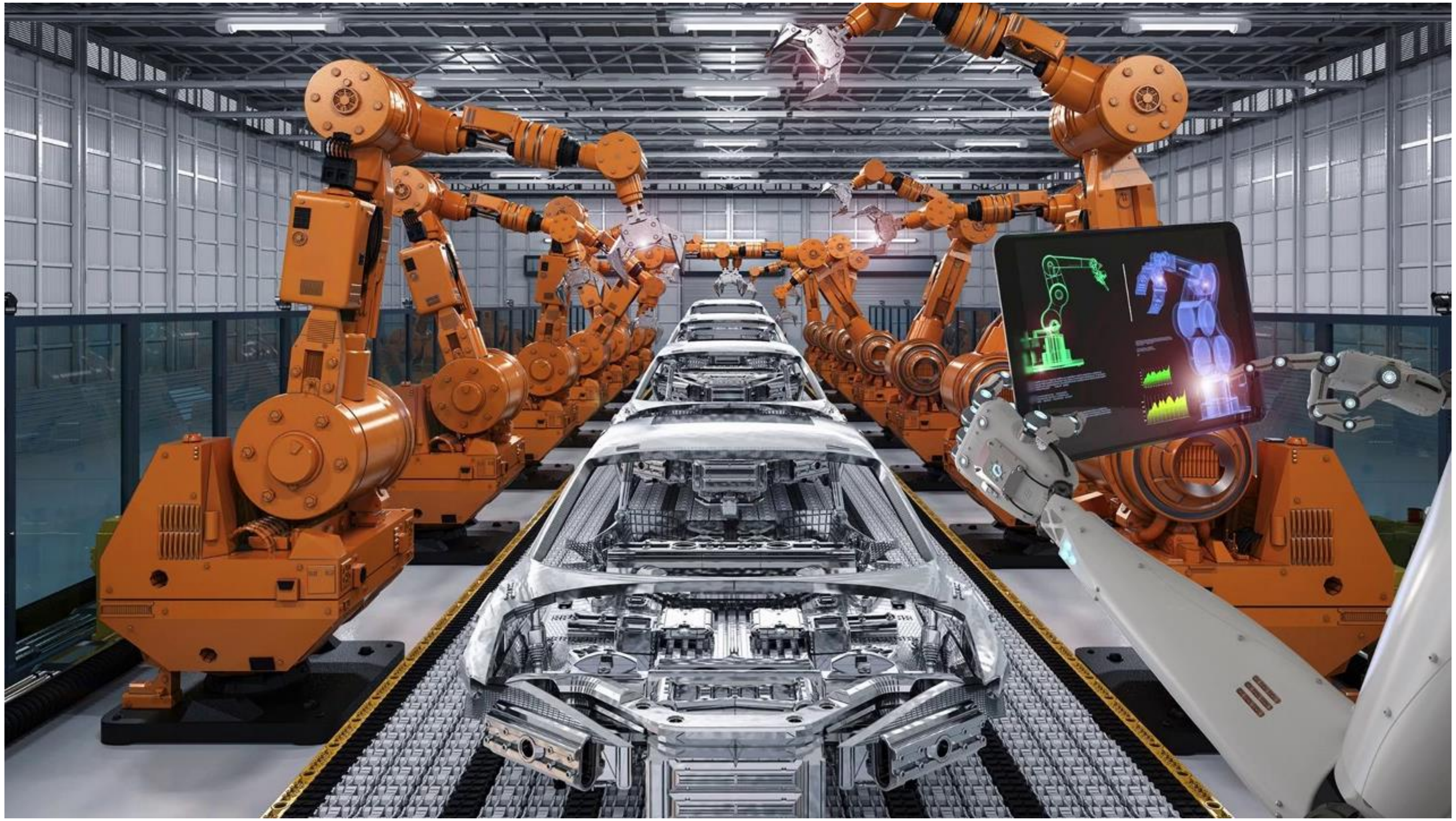
0. Anomaly detection

1. Motivations / Contributions

2. Proposed algorithm

3. Experiments

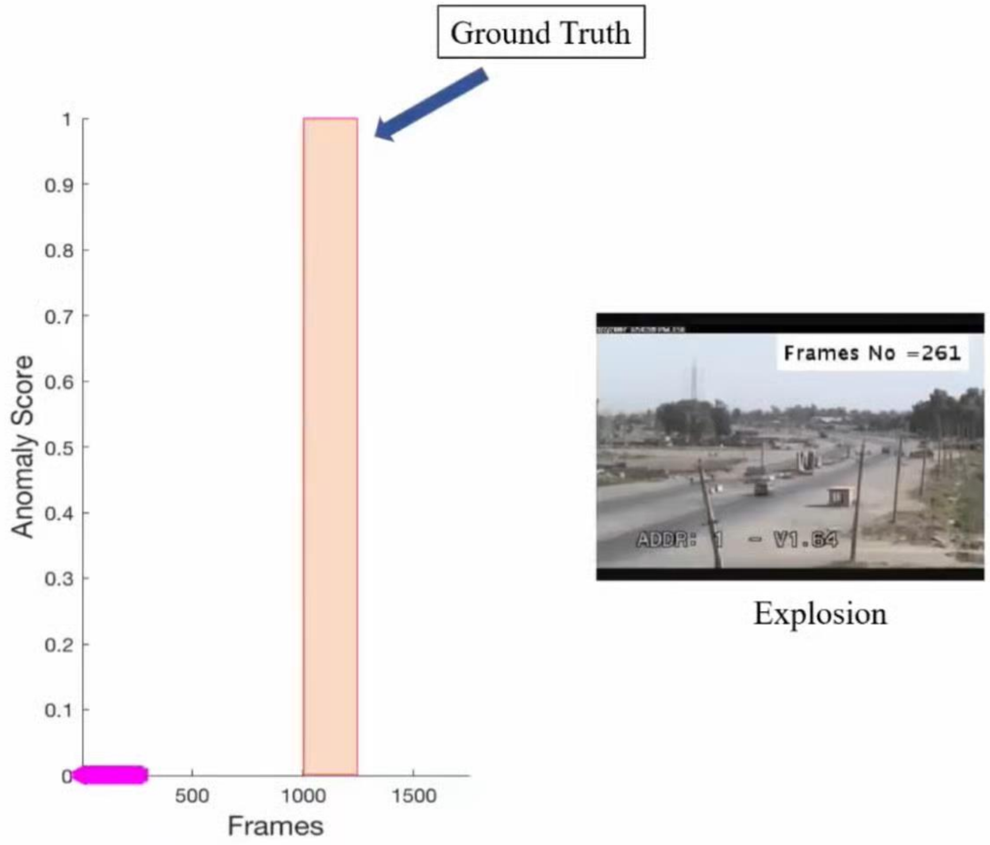
Anomaly detection



https://www.youtube.com/watch?v=mbbmeboRurI

Anomaly detection

- Use cases
  - Log Anomaly Detection
  - Fraud Detection
    - credit cards
  - Medical Anomaly Detection
    - MRI image
  - Industrial Anomaly Detection
    - robot arms
  - **Video Surveillance**



https://www.youtube.com/watch?v=qS1iPiSW3RE

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Motivation

- It is difficult to list all possible anomalous events.

- The boundary between normal and anomalous behaviors is often ambiguous.

Motivation

- Examples


http://www.svcl.ucsd.edu/projects/anomaly/dataset.html

https://www.merl.com/demos/video-anomaly-detection

Motivation

- The environment captured by surveillance cameras can change drastically over the time (e.g., at different times of a day), these approaches produce high false alarm rates for different normal behaviors.

- The same behavior could be a normal or an anomalous behavior under different conditions

Motivation

- Examples

https://www.kaggle.com/aalborguniversity/aau-rainsnow

How to solve?

- Anomaly detection should be done with minimum supervision.

An anomaly detection algorithm

using weakly labeled training videos

How to solve?

- Previous approach

0 0 0 0 1

- This paper

Video1: 1

Video2: 0

http://www.svcl.ucsd.edu/projects/anomaly/dataset.html

Contributions

- An anomaly detection algorithm using weakly labeled training videos
- A new large-scale video anomaly detection dataset consisting of 1900 real-world surveillance videos of 13 different anomalous events and normal activities captured by surveillance cameras.
- Superior performance as compared to the SOTA anomaly detection approaches (2018)

Any questions?

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Multiple Instance Learning [1, 2]

- Precise temporal locations of anomalous events in videos are unknown.
- Annotating them are laborious.

- Instead of receiving a set of instances which are individually labeled, **the learner receives a set of labeled bags, each containing many instances.**

https://en.wikipedia.org/wiki/Multiple_instance_learning

[1] T. G. Dietterich, R. H. Lathrop, and T. Lozano-P´erez. Solving the multiple instance problem with axis-parallel rectangles. Artificial Intelligence, 89(1):31–71, 1997.

[2] S. Andrews, I. Tsochantaridis, and T. Hofmann. Support vector machines for multiple-instance learning. In NIPS, pages 577–584, Cambridge, MA, USA, 2002. MIT Press.

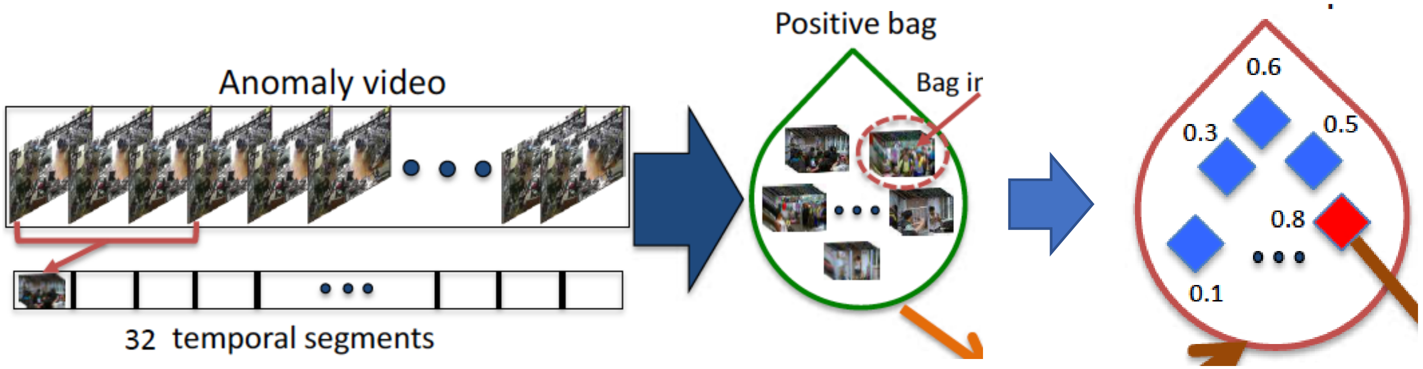
Multiple Instance Learning

In standard supervised classification problems using support vector machine, the labels of all positive and negative examples are available and the classifier is learned using the following optimization function:

$$\min_{\mathbf{w}} \quad \frac{1}{k} \sum_{i=1}^{k} \overbrace{max(0, 1 - y_i \boxed{(\mathbf{w}.\phi(x) - b)})}^{①} + \frac{1}{2} \left\|\mathbf{w}\right\|^2 , \qquad (1)$$

Multiple Instance Learning

- A positive bag $B_p = \{p^1, p^2, \dots, p^m\}$
- A negative bag $B_n = \{n^1, n^2, \dots, n^m\}$



$$\min_{\mathbf{w}} \frac{1}{z} \sum_{j=1}^{z} max(0, 1 - Y_{\mathcal{B}_j} \boxed{(\max_{i \in \mathcal{B}_j}(\mathbf{w}.\phi(x_i)) - b)}) + \frac{1}{2} \|\mathbf{w}\|^2, \quad (2)$$

Image from original paper

How to score each of them?

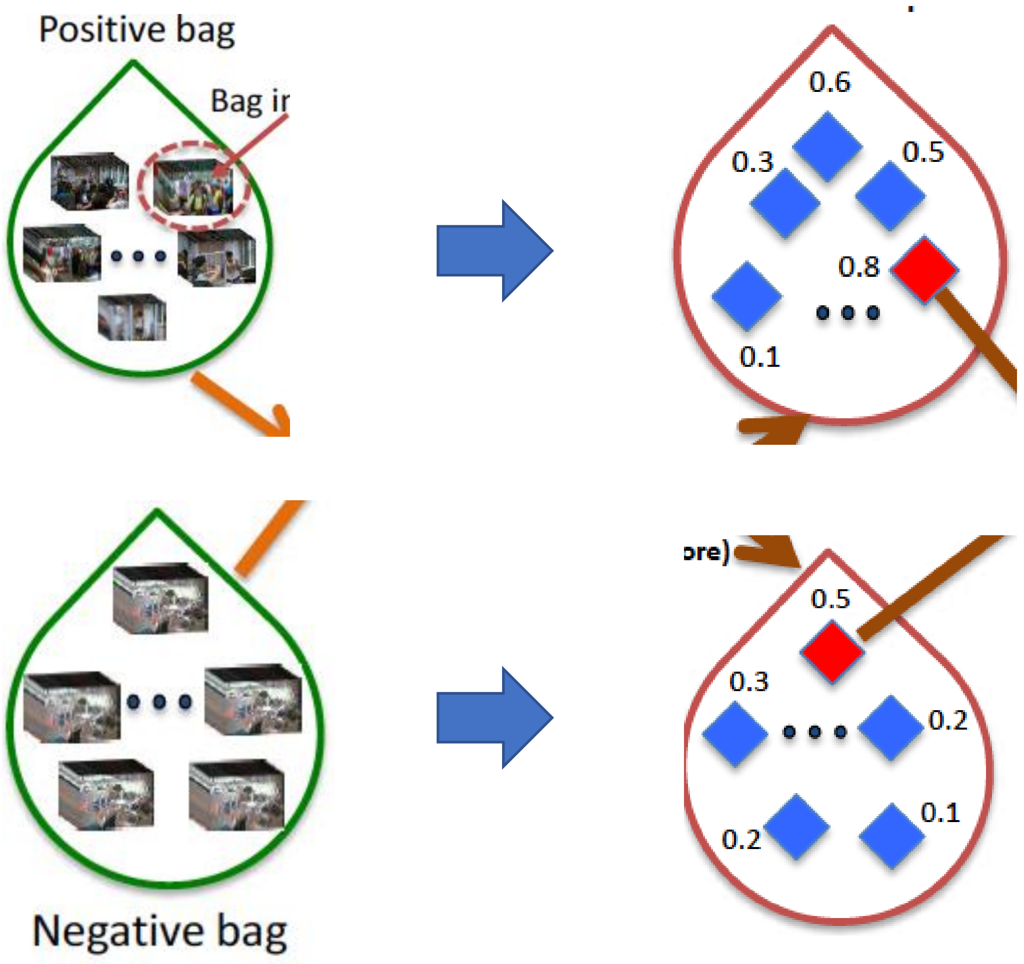
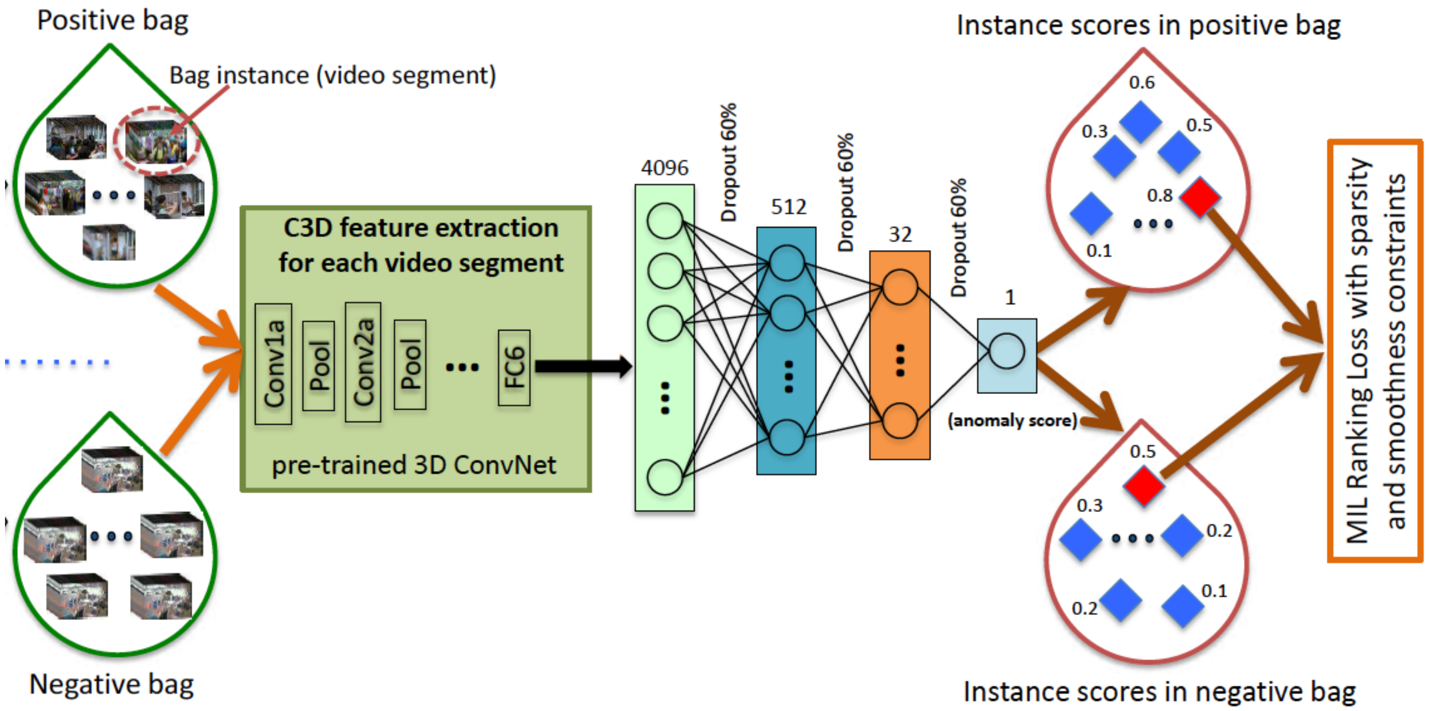


Image from original paper

Deep MIL Ranking Model

- Used C3D [1] pretrained model



[1] D. Tran, L. Bourdev, R. Fergus, L. Torresani, and M. Paluri. Learning spatiotemporal features with 3d convolutional networks. In ICCV, 2015.

Deep MIL Ranking Model

$$f(\mathcal{V}_a) > f(\mathcal{V}_n), \tag{3}$$

$$\max_{i \in \mathcal{B}_a} f(\mathcal{V}_a^i) > \max_{i \in \mathcal{B}_n} f(\mathcal{V}_n^i), \tag{4}$$

Deep MIL Ranking Model

- Limitation: it ignores the underlying temporal structure of the anomalous video
  - anomaly often occurs only for a short time.
  - since the video is a sequence of segments, the anomaly score should vary smoothly between video segments.
- temporal smoothness between anomaly scores of temporally adjacent video segments by minimizing the difference of scores for adjacent video segments.

Deep MIL Ranking Model

$$l(\mathcal{B}_a, \mathcal{B}_n) = \max(0, 1 - \max_{i \in \mathcal{B}_a} f(\mathcal{V}_a^i) + \max_{i \in \mathcal{B}_n} f(\mathcal{V}_n^i))$$

$$+\lambda_1 \overbrace{\sum_i^{(n-1)} (f(\mathcal{V}_a^i) - f(\mathcal{V}_a^{i+1}))^2}^{①} + \lambda_2 \overbrace{\sum_i^{n} f(\mathcal{V}_a^i)}^{②}, \qquad (6)$$

$\lambda_1 = \lambda_2 = 8 * 10^{-5}$

(the best performance)

Complete objective function

8). Finally, our complete objective function is given by

$$\mathcal{L}(\mathcal{W}) = l(\mathcal{B}_a, \mathcal{B}_n) + \lambda_3 \left\|\mathcal{W}\right\|_F , \quad (7)$$

where $\mathcal{W}$ represents model weights.

$\lambda_3 = 0.01$ (the best performance)

Overall pipeline

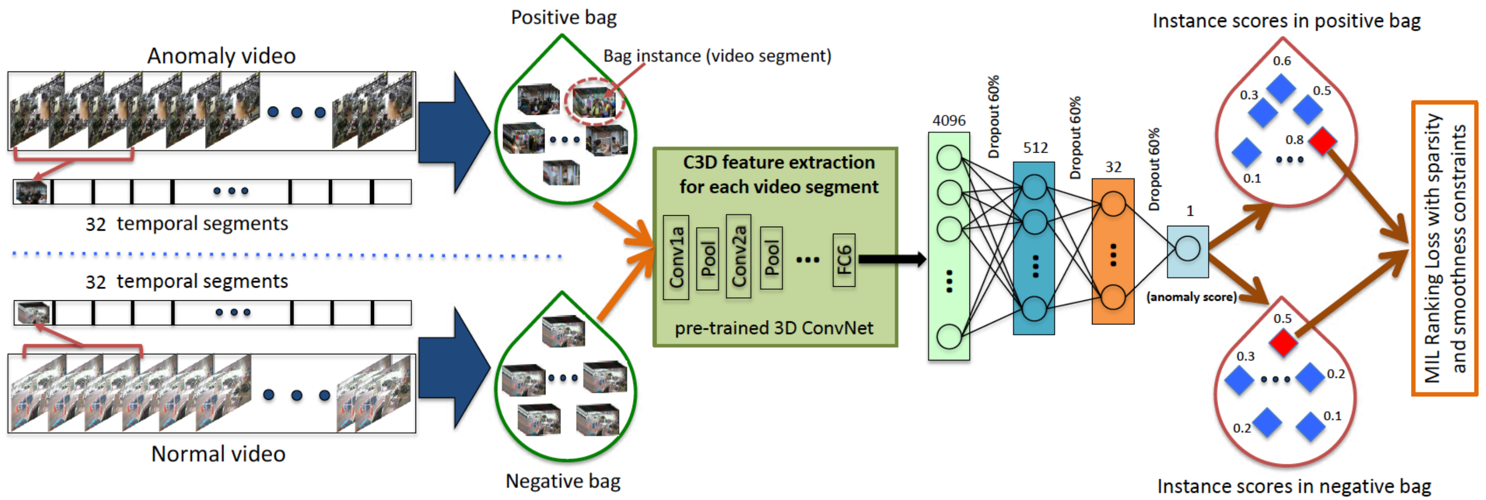


Figure 1. The flow diagram of the proposed anomaly detection approach. Given the positive (containing anomaly somewhere) and negative (containing no anomaly) videos, we divide each of them into multiple temporal video segments. Then, each video is represented as a bag and each temporal segment represents an instance in the bag. After extracting C3D features [37] for video segments, we train a fully connected neural network by utilizing a novel ranking loss function which computes the ranking loss between the highest scored instances (shown in red) in the positive bag and the negative bag.

Image from original paper

Any questions?

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Previous datasets

- UMN dataset
  - Anomaly: only running action
- UCSD ped1, ped2
- Avenue
- Subway exit, entrance
- BOSS
- Abnormal crowd

Proposed dataset

- long untrimmed surveillance videos which cover 13 real-world anomalies, including Abuse, Arrest, Arson, Assault, Accident, Burglary, Explosion, Fighting, Robbery, Shooting, Stealing, Shoplifting, and Vandalism.

UCF-crime dataset

https://www.crcv.ucf.edu/research/real-world-anomaly-detection-in-surveillance-videos/

Datasets (table)

	# of videos	Average # of frames	Dataset length	Example anomalies
UCSD Ped1 [27]	70	201	5 min	Bikers, small carts, walking across walkways
UCSD Ped2 [27]	28	163	5 min	Bikers, small carts, walking across walkways
Subway Entrance [3]	1	121,749	1.5 hours	Wrong direction, No payment
Subwa Exit [3]	1	64,901	1.5 hours	Wrong direction, No payment
Avenue [28]	37	839	30 min	Run, throw, new object
UMN [2]	5	1290	5 min	Run
BOSS [1]	12	4052	27 min	Harass, disease, panic
Abnormal Crowd [31]	31	1408	24 min	Panic, fight, congestion, obstacle, neutral
Ours	**1900**	**7247**	**128 hours**	**Abuse, arrest, arson, assault, accident, burglary, fighting, robbery**

Table 1. A comparison of anomaly datasets. Our dataset contains larger number of longer surveillance videos with more realistic anomalies.

Baseline methods

- Lu et al. ([1], dictionary based approach)
- Hasan et al. ([2], a fully convolutional feedforward deep auto-encoder based approach)
- Binary SVM classifier

[1] C. Lu, J. Shi, and J. Jia. Abnormal event detection at 150 fps in matlab. In ICCV, 2013.
[2] M. Hasan, J. Choi, J. Neumann, A. K. Roy-Chowdhury, and L. S. Davis. Learning temporal regularity in video sequences. In CVPR, June 2016.

Evaluation metrics

- ROC curve

- AUC (area under curve)

Quantitative results

Method	AUC
Binary classifier	50.0
Hasan *et al*. [18]	50.6
Lu *et al*. [28]	65.51
Proposed w/o constraints	74.44
Proposed w constraints	**75.41**

Table 3. AUC comparison of various approaches on our dataset.

Method	[18]	[28]	**Proposed**
False alarm rate	27.2	3.1	**1.9**

Table 4. False alarm rate comparison on normal testing videos.

Image from original paper

Quantitative results

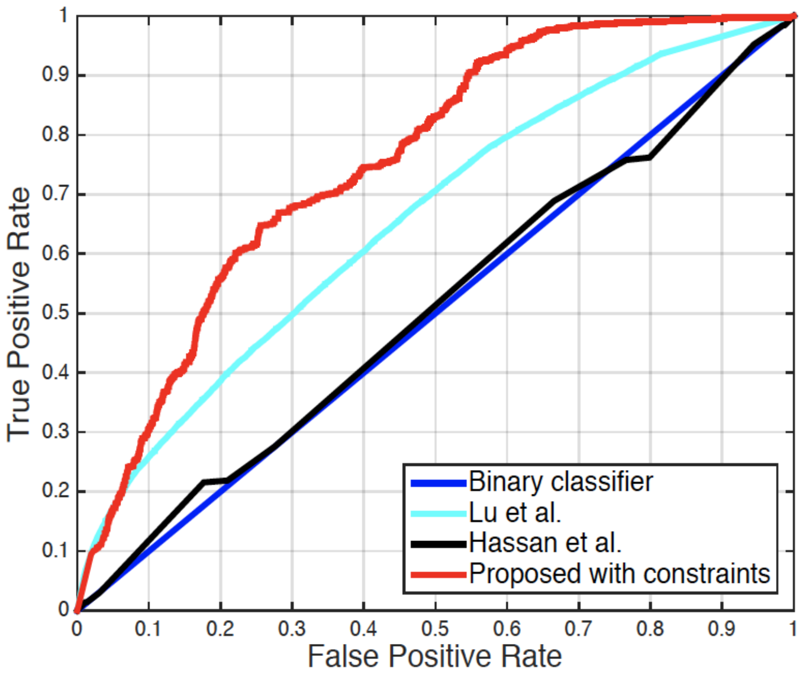

Figure 6. ROC comparison of binary classifier (blue), Lu *et al.* [28] (cyan), Hasan *et al.* [18] (black), proposed method without constraints (magenta) and with constraints (red).

Image from original paper

Qualitative results

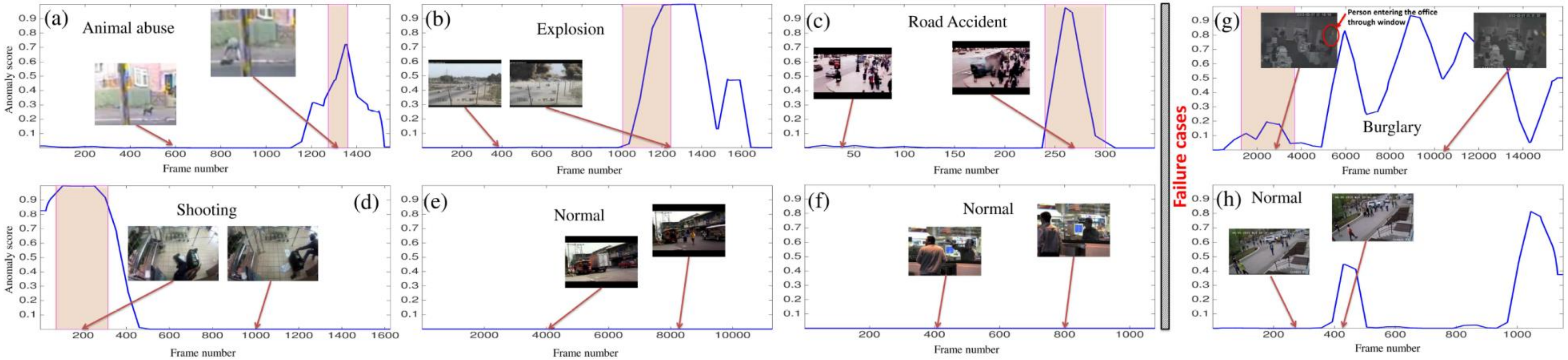

Figure 7. Qualitative results of our method on testing videos. Colored window shows ground truth anomalous region. (a), (b), (c) and (d) show videos containing *animal abuse (beating a dog)*, *explosion*, *road accident* and *shooting*, respectively. (e) and (f) show normal videos with no anomaly. (g) and (h) present two failure cases of our anomaly detection method.

Image from original paper

Conclusion

- Proposed a deep learning approach to detect real-world anomalies in surveillance videos
- Introduced a new large-scale anomaly dataset consisting of a variety of real-world anomalies
- Outperformed previous methods

Any questions?